

COURTSIDE: Pricing the Value of Speed in In-Play Tennis Prediction Markets

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Abstract. In-play tennis prediction markets are priced by whoever learns the point first. We measure who that is, what they earn and what each second is worth, from public trade tapes of 13,084 Polymarket tennis matches. A walk-forward fast tier of wallets trading within 3 s of a score move earns +0.93¢ per share after fees in sample and +0.75¢ in its single out-of-sample run; other takers lose. At its own fills a frozen book earns +0.60¢ on a burned (non-blind) hold-out, but not with fees doubled. Simulated at a 1 s licensed-feed baseline (assumed feed latency (licensed feed not purchased); parameters measured), a computer-vision (CV) trader earns +\$4 a day on that hold-out at the pre-registered stamp lag, where each second of feed delay costs \$42–74 a day, and +\$57 at a post hoc estimate, on trades selected on outcomes; a replay on real books loses. Real money: none.

JEL: G14, G12, C58. **Keywords:** prediction markets; latency arbitrage; in-play betting; tennis; computer vision; market microstructure.

1 Summary

Our CV calls the point before the ball lands; we price the feed it needs.

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- **Strategy and edge.** Call the point with CV; lift the stale resting quote inside the 1 s taker delay. The book reprices 1.2 s *before* the official stamp; the fast tier earns +0.93 / +0.75¢ a share after fees (IS / OOS) every month; other takers lose.
 - **Headline, burned OOS.** v2 at fast-tier fills: +0.60¢ [0.09, 1.13], Sharpe 6.7. CV at the 1 s baseline (assumed feed latency (licensed feed not purchased); parameters measured): +\$4/day (Sharpe 0.3) at the pre-registered $L = 2.0$ s, +\$57/day (Sharpe 8.8) at the post hoc $L = 3.14$ s [2.23–3.22]; trades selected on outcomes, not ex ante.
 - **Failed; decision.** Fees $\times 2$ OOS (–0.34¢); every blind test of a tradable book; the replay on real books; table tennis. Go only on a measured stamp lag $L \geq 3.0$ s, which pays for the cheapest data stack. Real money: none.
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2 Economic hypothesis

Win probability is a known function of the score. The leverage of point k ,

$$\Delta(k) = \Pr(\text{win} \mid \text{point } k \text{ won}) - \Pr(\text{win} \mid \text{point } k \text{ lost}), \quad (1)$$

follows from the exact point-level Markov chain (Klaassen and Magnus, 2001), with mean $|\Delta(k)|$ of 5.7% over the 161 points of a simulated ATP best-of-three. Each point moves fair value by a known amount at a known instant, and the news reaches traders in tiers (Fig. A1a): in-venue tracking; the official stamp (unmeasured, 1–3 s later); licensed video (0.5–8 s, vendor-stated); public streams (≥ 12.2 s); score feeds (28.2–44.1 s after the book, which reprices a median 1.2 s *before* the stamp,

Reproduce: bash reproduce.sh (<https://github.com/OjasMishra32/computervisionpredictivetabletennis>). Paper only; no licensed feed, match footage or court camera is held.

$n = 482$). Makers whose quotes are stale when a point ends are picked off by whoever learns it first: the adverse selection of [Glosten and Milgrom \(1985\)](#), the stale-quote sniping of [Budish et al. \(2015\)](#) and [Aquilina et al. \(2022\)](#), the toxic arbitrage of [Foucault et al. \(2017\)](#).

H6. *Universe:* Polymarket ATP/WTa/Challenger moneylines. *Behaviour:* wallets trading 0–3 s after a score move. *Horizon:* 30 s markout. *Counterparty:* the resting maker who has not yet cancelled a stale quote. *Persistence:* physics and data licences decide who learns first; the 1 s taker delay lets makers cancel, so only news that leads the reprice by over a second pays (a barrier to entry; the cost D in Eq. 2). *Prediction:* fast wallets chosen on past months keep a positive net markout each month and out of sample. *Falsifier:* ≤ 0 in over a third of months, or out of sample. *Trail:* H1–H4 committed before any result (7232986); H6 frozen before the OOS opened; changes in DEVIATIONS.md.

The rent tracks the structure (v2: +1.96, +1.36, +1.45, +1.02¢ in the 3 s/0%, 3 s/3%, 1 s/3%, 1 s/5% delay/fee regimes) and is being competed away: the fast tier grew from 4 to 131 wallets as its edge fell 0.20¢ a month ($t = -3.98$); five wallets carry 81% / 142% of v2's P&L; the venue now buys official data and streams the ATP itself ([Gaming Intelligence, 2026](#); [PR Newswire, 2026](#)). In-play mispricing is known ([Croxson and Reade, 2014](#); [Brown, 2012, 2014](#)); new here: a wallet-level map of the latency tier, a timing of every public source, a price on each second.

3 Data and universe

13,084 resolved ATP/WTa/Challenger singles moneylines with $\geq \$5,000$ volume, Oct 8, 2025 to Oct 3, 2026, on a public order book. IS: the first 80% of matches by start time; OOS: the last 2,617 from 2026-08-25 14:15 UTC (21.2% of volume; burned for v2 and CV). A blind test on 11,307 smaller markets checks survivorship. 6 tapes may be truncated, 2,889 early matches pay no fee, 2.89% settle 50/50 (all kept); tape stamps lag 1.98 s. Sources: Polymarket APIs and sports websocket ([Polymarket, 2026a](#)); ESPN and WTA, latency only ([ESPN, 2026](#); [WTA, 2026](#)); Kalshi ([Kalshi, 2026](#)); [French \(2026\)](#); OpenTTGames ([Voeikov et al., 2020](#)); BlurBall and WASB-SBDT weights ([Gossard et al., 2026](#); [Tarashima et al., 2023](#)).

4 Methodology

Fast tier and v2. A detector fires on a ≥ 4 ¢ move of the 10 s VWAP against the prior 60 s. Each month a wallet qualifies on past months only (≥ 30 prints 0–3 s after a detection, ≥ 10 matches, $t > 3$ on its 30 s markout). The frozen book v2 copies qualifying prints, sized $\propto 1/\sqrt{q(1-q)}$, at most \$1,000 per order and 100 shares net per match, held to resolution, at the fast tier's own prices: *the opportunity at their speed, not our execution*.

CV strategy. Streamed causally on held-out 120 fps table-tennis video, our live engine calls 4 of 41 misses before contact, none wrongly (Wilson lower bound 51%; median lead 162 ms); an offline evaluation with a look-ahead feature called 11 of 11 at 50 ms (Fig. A1b). Frame to order-ready over WebRTC takes 54 ms; with a simulated 1 s feed, network and venue hold, 2,119 ms, under the organiser's 3 s bar (results/e2e, paper). With the bounce at time zero an order wins only if

$$V + \delta + D - \ell < L + R \iff V < L + R - D - \delta + \ell, \quad (2)$$

where V is the feed delay, δ the inference (20 ms), network and gateway, D the 1 s taker delay, ℓ the call lead, L the stamp lag and R the reprice relative to the stamp, measured on 482 live points (fills draw on the 265 whose book moved ≥ 3 ¢); V runs 0–60 s, 20 seeds a cell. The trade set is every historical point the market later repriced ≥ 4 ¢: *selected on outcomes, not ex ante*. L is unmeasured: pre-registered 2.0 s, or a post hoc estimate of 3.14 s (95% CI 2.23–3.22 s; two modes, as the stamp has 1 s resolution) that assumes the first informed prints are courtside humans reacting in 0.25 s (who would also queue ahead of a 1 s-feed order). Free of L , the call must land ≥ 0.89 s before the umpire's stamp; $V = 1$ s at $L = 3.14$ s equals a camera 0.14 s *ahead* of the bounce at $L = 2.0$ s. Its pre-registration set no pass/fail rule; the blind test of the frozen v3 rule did, and failed (Table 3).

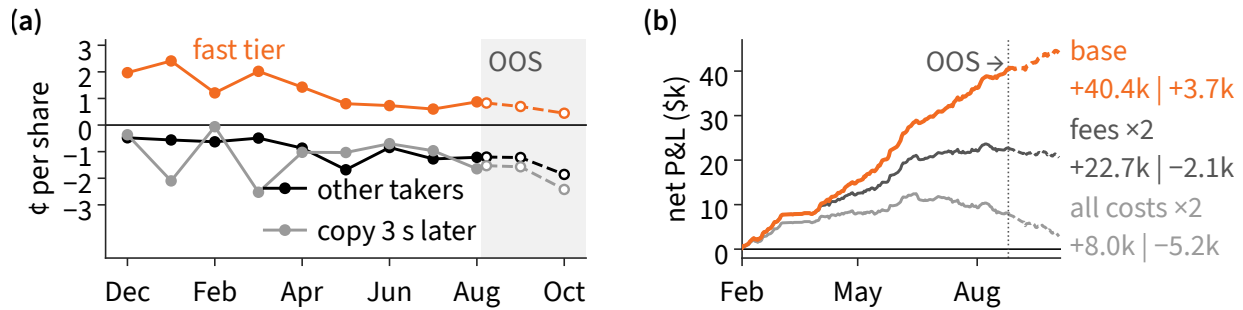


Figure 1. *Speed is the edge.* (a) Monthly net 30 s markout after fees; filled = IS, hollow = OOS. (b) v2 net P&L at the fast tier's own fills, base and costs doubled; solid = IS, dashed = burned OOS; labels: IS | OOS totals. Source: results/{alpha,lowloss,v2}.

Table 1. v2 earns in both periods at the fast tier's own fills; its OOS edge does not survive doubled fees.

	In sample Feb 1–Aug 25, 206 d	Burned OOS (non-blind) Aug 25–Oct 3, 40 d
Net ¢ per share [95% CI]	+1.38 [1.17, 1.59]	+0.60 [0.09, 1.13]
Annualised return / volatility	253% / 17.5%	148% / 22.2%
Sharpe, annualised [bootstrap 95% CI]	14.5 [11.9, 17.4]	6.7 [1.9, 12.3]
Max drawdown / worst day, % of capital	−2.0% / −1.9%	−2.1% / −2.1%
Skew / worst month	+0.53 / +\$2,543	+0.33 / −\$434
Turnover (× capital a year) / months positive	93 / 7/7	129 / 2/3
Costs stressed: ½ / 1 tick worse entry	+0.88 / +0.38	+0.10 / −0.40
Fees ×2 [CI] (months positive)	+0.77 [0.57, 0.98] (7/7)	−0.34 [−0.86, 0.19] (0/3)
All costs ×2 [CI]	+0.27 [0.07, 0.48]	−0.84 [−1.36, −0.31]

Notes: 55,662 / 10,412 trades in 7,639 / 2,003 matches, held to resolution, net of fees and spreads (net edge 273 / 114 bps of notional); capital 3× peak locked (\$28,302 / \$22,754); CIs by match (by wallet, OOS: [−0.45, 2.38]). Source: results/v2/.

Costs. Each match's own taker fee,

$$f(q) = r q (1 - q) \text{ per share} = 10,000 r (1 - q) \text{ bps of notional}, \quad (3)$$

with $r = 5\%$ today (250 bps at $q = 0.5$): 120 / 179 bps on v2's trades (IS / OOS). Spreads are paid as traded. Stresses: ½ and 1 tick worse entry, fees ×2, all costs ×2 (≈+99 bps more). **Validation.** Walk-forward inside IS; the fee in force; the onset look-ahead fixed (D9); every read of held-out data logged.

5 Results

The fast tier beats other takers every month and copying it 3 s later loses (Fig. 1a). At its fills v2 earns in both periods (Table 1, Fig. 1b); out of sample the edge is gone at ½ tick worse entry (+0.10¢) and negative with fees doubled (−0.34¢, 0/3 months positive). **Sharpe above 3.** We assumed a bug and found one: fixing the onset look-ahead (D9) cut IS Sharpe from 16.8 to 14.5. The rest is about 270 small, nearly independent bets a day (same-day correlation 0.0013). The deflated Sharpe (Bailey and López de Prado, 2014) is 0.997 IS at $N = 3,386$ but 0.075–0.839 OOS: 40 days cannot rule out luck. The CV Sharpe is a seed mean over trades selected on outcomes. **The 1 s baseline.** At the pre-registered lag the CV trader makes +\$15 / +\$4 a day (IS / OOS; Sharpe 2.1 / 0.3); at the post hoc estimate +\$94 / +\$57 (Sharpe 11.9 / 8.8; Table 2, Fig. 2). That figure is fragile: at its interval's low end it is about +\$19 / +\$8, and the same inference read per point falls to −\$17 (break-even 0.34 / 0.29 s). The live engine's own calls barely move the cells. Two biases flatter the sweep: trades chosen on outcomes, and a reprice timed 66 ms late. The tests free of them lose: a replay of 9 matches

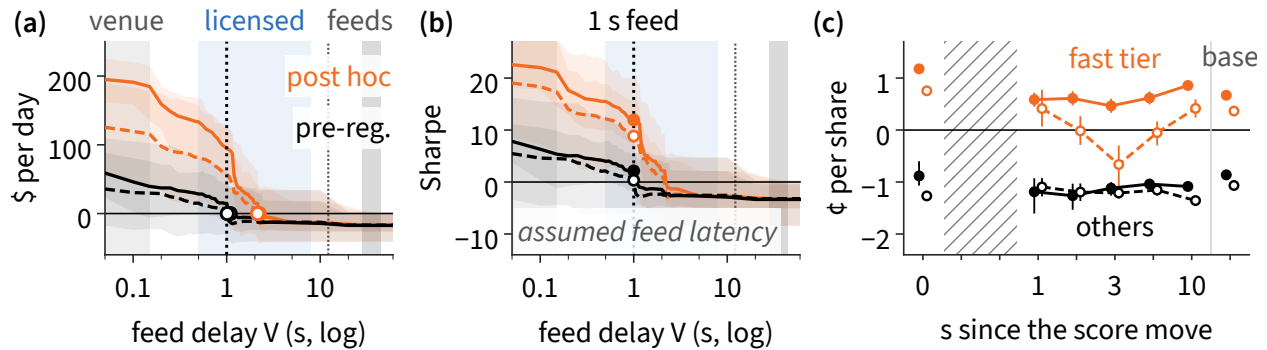


Figure 2. *Signal decay: each second of feed latency costs the edge.* (a) \$/day and (b) Sharpe against feed delay V at the pre-registered (black) and post hoc (orange) stamp lag; solid = IS, dashed = burned OOS; bands: 20 seeds; circles: break-even; shading: venue, licensed video, score feeds; dotted: 1 s, public stream. (c) Net markout by seconds since the move. Simulated: assumed feed latency (licensed feed not purchased); parameters measured; trades selected on outcomes. Source: results/{tier0,decay}.

Table 2. *Main result: the CV trader simulated at a 1 s licensed-feed baseline; assumed feed latency (licensed feed not purchased); parameters measured; trade set: historical points the market later repriced $\geq 4\text{¢}$ (selected on outcomes, not ex ante).*

At $V = 1\text{ s}$	Pre-registered $L = 2.0\text{ s}$		Post hoc estimate $L = 3.14\text{ s}$	
	IS	OOS	IS	OOS
Net \$ per day [seed 95%]	+\$15 [-9, 43]	+\$4 [-36, 62]	+\$94 [51, 125]	+\$57 [1, 130]
Net ¢ per share	+0.40	-0.38	+1.11	+0.66
95% CI	[-0.32, 1.08]	[-2.08, 1.21]	[0.85, 1.38]	[0.10, 1.19]
Return / vol, % a year / Sharpe	30 / 14.1 / 2.1	11 / 33.4 / 0.3	118 / 9.9 / 11.9	91 / 10.3 / 8.8
Max DD % / worst day	-6.6 / -\$263	-5.3 / -\$228	-1.6 / -\$223	-1.6 / -\$192
Wrong calls: share / P&L	35% / -\$2,403	47% / -\$938	12% / -\$4,141	16% / -\$1,270
Break-even feed delay V , s	1.09	1.01	2.23	2.14
\$/day, live engine's calls	+\$17	+\$4	+\$98	+\$57
\$/day gained, $V: 1\text{ s} \rightarrow 0$	+\$74	+\$42	+\$104	+\$73

Notes: 20 draws; 10 matches a day; IS 103 d, OOS 40 d (burned); 24–65 trades a day; no daily path (equity curve: v2). Post hoc L : 95% CI 2.23–3.22 s. At $L = 1\text{ s}$: -\$6 / -\$11 a day. Pre-registered: PREREG.md, written 17:25 UTC before the first P&L, committed with the results (a5769c7, 19:24); timing per tournament from V3.

recorded live on 2026-10-03 on their real books loses in 36 of 36 settings (-0.94¢ at $V = 1\text{ s}$; Fig. A3), and an ex-ante Markov-swing filter (exploratory) in 9 of 9 cells (Table A3), so the feed delay, not the selection, sinks the 1 s trader; the frozen v3 rule failed blind (Table 3).

6 Risk management

- **Limits, in code** (engine/risk/). \$1,000 per order, 100 shares net per match, zone 0.05–0.95: one match can lose at most \$95 plus fees. Capital 3× peak locked (v2 \$28,302; CV \$13,800–\$29,230). Kill switches: stale feed >2 s, vision >1 s, latency above its rolling p95, manual; the \$1,000 daily stop (3.86 σ of v2's daily P&L) is a disaster stop.
- **CV model risk.** Wrong calls always fill (Table 2). MISS calls trade only at confidence ≥ 0.95 and positive edge after the fee (code); live use needs 73 audited correct calls in a row, and a confirmed false call halts the day (policy). Phantom calls outside play (about 30 an hour) erase the 1 s OOS

Table 3. *Robustness: every blind or ex-ante test of a tradable book failed.*

Test, ¢ per share	IS	OOS or blind	Verdict
H1 · H2 · H5 (post hoc)	-1.61 · -0.41 · +0.22	-2.08 · +0.80 · +0.16	fail · none · n.s.
v1 copy fast tier (OOS opened once)	+1.16	+0.57 (-\$36,056)	fail
v2, 11,307 unseen markets (blind)	+2.02	+1.22 [-0.19, 2.65]	fail
CV v3 rule, unseen markets (blind)	-0.22	-0.52	fail
Maker v1 (blind), per fill	+2.02	+1.87 [-0.14, 3.86]	fail, -\$379
v2 after fixed costs, \$ a day	+\$29	-\$75	fail OOS
Table tennis TT1–TT5, 27 matches	no fast wallet qualifies		untestable
CV replay on live books, $V = 1$ s		-0.94 [-1.28, -0.58]	loses
ex-ante swing $\geq 4\text{¢}$ (exploratory)		-0.95 [-1.64, -0.40]	loses

Notes: **4,219 variants tried** (Table A5). Fast minus others, unseen markets: +3.09 / +2.16¢. Held-out data read 76 times (7 blind first runs, 16 non-blind, 5 descriptive, 25 audits, 21 replay/live, 2 other; Table A8); rules changed after a look: 2 for v2, 13 for the CV simulation (T1–T3, V1–V10).

P&L at 2.5 (pre-registered) or 32 (post hoc) a day, so the rally-state gate (coded, off, unevaluated) must be on. Betting video at 25–50 fps without early calls gives +\$8 / -\$13 a day (pre-registered).

- **De-risking, set in advance** (policy, not code; it never fired in sample, so it is untested on a decline). Halve size if the trailing 30-day edge is below 0.3¢ (IS minimum +0.74¢); stop at ≤ 0 , a 5% drawdown, or a fee or delay change. At the fast tier's decay, v2's OOS +0.60¢ halves size in 1.5 months and stops in 3.0. The CV book starts only on a measured $L \geq 3.0$ s.
- **Tail, venue, legal.** Fees $\times 2$ erase v2's OOS edge; kurtosis 4.30 / 3.26. The UMA oracle settles (disputes: 4–6 days); collateral is pUSD, so only working capital sits on the venue (Polymarket, 2026c); US persons are close-only (Polymarket, 2026b); courtsiding breaches ticket terms (International Tennis Federation, 2026). Factors (Fama and French, 1993; Carhart, 1997): α $t = 8.48$, $\max |t| = 1.36$, R^2 3.3% (IS; Table A4).

7 Liquidity and capacity

v2 trades \$7,182 a day, 0.07% of match volume; its OOS CI excludes zero only up to 1 $\times$ size (\$22,754) and 5 $\times$ loses (Fig. A2a). Median stale depth before a reprice: \$222 / \$379 / \$564 at 0 / 1 / 2 s; spread 1¢. The CV book (Table 2: 10 matches a day, half the stale depth) at the post hoc estimate halves its Sharpe at \$41,000 of capital OOS (\$66/day) and \$74,000 IS (\$181/day), or \$96,000 / \$164,000 covering every match; at the pre-registered lag its Sharpe is 2.7 / 0.9 at the smallest size: no capacity. Its notional is 0.15% of in-play volume but 6% of the fast tier's 0–3 s volume: the seconds after each point bind. Fixed costs decide: the book can pay at most \$1,644 a month for data OOS (post hoc) or \$55 (pre-registered), against an assumed licence of \$1,250–\$10,000; the central \$167-a-day stack is covered at no $L \leq 3.0$ s, the cheapest (\$42) only at $L \geq 2.87$ –2.98 s. Covering every match, the camera-free stack (\$242–\$256 a day) breaks even from about \$52,000 (IS) and \$99,000 (OOS) of capital (results/capacity); v2's OOS break-even fee is 8.2% (Fig. A2b).

8 Limitations and next steps

We hold a cheap real option on one number, the stamp lag L , which moves the break-even one for one (Eq. 2). A live session with a licensed low-latency feed and small test orders measures it, and whether delayed orders are released on a 1 s clock (35% of reprices land just after a whole second); go if $L \geq 3.0$ s. Licensing turns the simulation into deployment (Polymarket itself buys official data; Gaming Intelligence, 2026); a faster feed moves left on the curve (Fig. 2a). CV accuracy is from table tennis (illiquid: spread 94¢); tennis is simulated (Fig. A8). Forward test: pending (runs once, Oct 4); live paper session: pending (paper; 0 fills so far).

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Disclosure. No real money was used. Simulated components are labelled. The authors hold no licensed data feed, no match footage and no court access. University of Florida students; [team to confirm any other disclosures].

Appendix

Everything a judge needs for scoring is in the five main pages; this appendix gives the full grids, logs and tallies behind them. Every appendix float repeats its label.

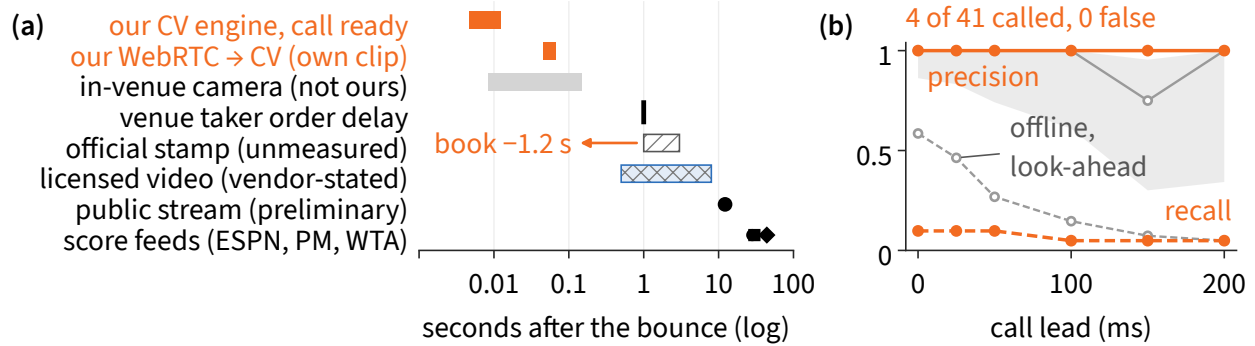


Figure A1. The race is decided inside the venue's 1 s taker delay; only in-venue information beats the book. (a) Delay after the bounce (log): orange = ours (WebRTC: our own clip, loopback, 46 ms p50, 67 ms p99); hatched = vendor-stated or model range. (b) Early out-calls on held-out 120 fps table-tennis video (41 out, 130 in): the live causal engine (orange; 4 of 41 misses called, no false call) against the offline evaluation with a look-ahead feature (grey; band: Wilson 95%). Source: Polymarket and WTA feeds; results/{tracking,engine,webrtc}.

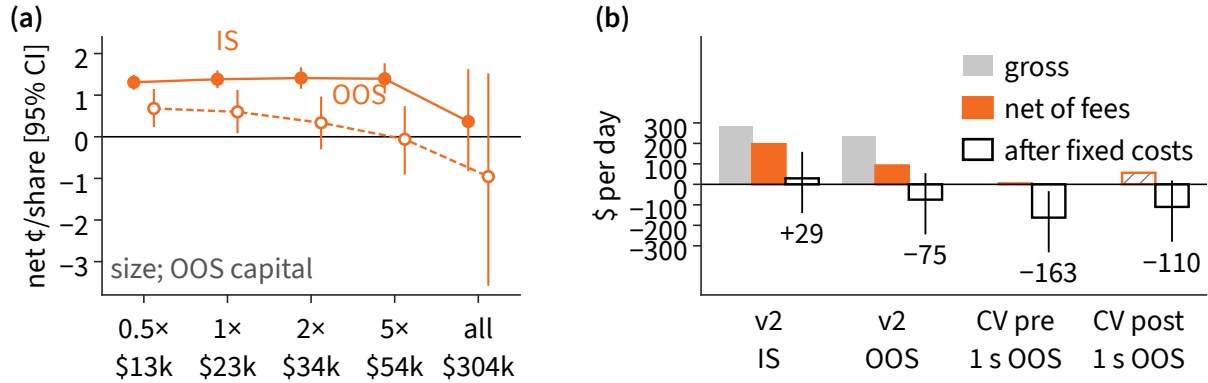


Figure A2. Capacity is small and costs bind. (a) v2 net edge by size (caps scaled; optimistic at large size); filled = IS, hollow = OOS. (b) \$/day: gross, net of fees, after fixed costs (low-high); CV pre / post: the 1 s baseline, burned OOS, at the pre-registered / post hoc stamp lag; assumed feed latency (licensed feed not purchased); parameters measured. Source: results/financials.

Table A1. Latency sweep, every reading: Net \$ per day against feed delay V (s), IS and burned OOS. Simulated: assumed feed latency (licensed feed not purchased); parameters measured.

Stamp lag reading		0	0.5	1	2	5	10	60
pre-registered 2.0 s	IS	89	28	15	-6	-13	-14	-17
	OOS	46	15	4	-11	-13	-13	-17
post hoc estimate 3.14 s	IS	199	133	94	15	-13	-13	-17
	OOS	129	81	57	7	-13	-13	-17
stress 1.0 s	IS	15	-5	-6	-13	-13	-14	-17
	OOS	4	-11	-11	-13	-13	-13	-17
lag 3.0 s	IS	189	121	89	15	-13	-13	-17
	OOS	117	76	46	4	-13	-13	-17
per point, 2.0 s	IS	-30	-17	-17	-17	-17	-17	-17
	OOS	-26	-17	-17	-17	-17	-17	-17
per point, post hoc	IS	199	-45	-17	-17	-17	-17	-17
	OOS	133	-40	-17	-17	-17	-17	-17

Notes: seed means over 20 draws per cell (18,880 runs in all); trade set selected on outcomes, not ex ante. The per-point readings treat the spread of the reprice time as umpire-stamp noise, so the reprice comes a fixed time after the bounce: at the pre-registered lag, and at the post hoc inference applied point by point (which wins only below 0.34 s). Source: results/tier0/latency_sweep.json::video_own120.

Table A2. Latency sweep, every reading: Annualised Sharpe against feed delay V (s), IS and burned OOS. Simulated: assumed feed latency (licensed feed not purchased); parameters measured.

Stamp lag reading		0	0.5	1	2	5	10	60
pre-registered 2.0 s	IS	11.4	4.0	2.1	-1.4	-2.7	-2.9	-3.4
	OOS	7.3	2.1	0.3	-2.3	-2.6	-2.6	-3.2
post hoc estimate 3.14 s	IS	22.9	15.5	11.9	2.2	-2.7	-2.7	-3.4
	OOS	19.6	12.2	8.8	0.9	-2.6	-2.6	-3.2
stress 1.0 s	IS	2.1	-1.3	-1.4	-2.6	-2.7	-2.9	-3.4
	OOS	0.3	-2.3	-2.3	-2.6	-2.6	-2.6	-3.2
lag 3.0 s	IS	21.8	14.4	11.4	2.1	-2.7	-2.7	-3.4
	OOS	17.5	11.4	7.3	0.3	-2.6	-2.6	-3.2
per point, 2.0 s	IS	-5.9	-3.4	-3.4	-3.4	-3.4	-3.4	-3.4
	OOS	-5.0	-3.2	-3.2	-3.2	-3.2	-3.2	-3.2
per point, post hoc	IS	25.1	-8.7	-3.4	-3.4	-3.4	-3.4	-3.4
	OOS	21.2	-7.6	-3.2	-3.2	-3.2	-3.2	-3.2

Notes: seed means over 20 draws per cell (18,880 runs in all); trade set selected on outcomes, not ex ante. The per-point readings treat the spread of the reprice time as umpire-stamp noise, so the reprice comes a fixed time after the bounce: at the pre-registered lag, and at the post hoc inference applied point by point (which wins only below 0.34 s). Source: results/tier0/latency_sweep.json::video_own120.

Table A3. Replay on 9 matches' real books (2026-10-03): calling every point or only an ex-ante Markov swing $\geq T$ loses at every V at the pre-registered lag. Net ¢ per share marked at +30 s [match-clustered 95% CI] (fills). Exploratory: the selective variant was added after the all-points replay; one day; assumed feed latency.

Lag	Points traded	$V = 0$	$V = 0.5$ s	$V = 1$ s
2.0 s	every point	-0.61 [-0.90, -0.38] (172)	-0.66 [-0.96, -0.41] (167)	-0.94 [-1.28, -0.58] (149)
	swing $\geq 2\text{¢}$	-0.55 [-0.91, -0.22] (140)	-0.61 [-1.02, -0.23] (134)	-0.92 [-1.35, -0.49] (114)
	swing $\geq 4\text{¢}$	-0.37 [-0.75, 0.11] (88)	-0.50 [-0.93, -0.10] (81)	-0.95 [-1.64, -0.40] (63)
	swing $\geq 6\text{¢}$	-0.36 [-0.99, -0.09] (48)	-0.22 [-0.91, 0.20] (47)	-0.26 [-1.35, 0.53] (33)
3.0 s	every point	-0.05 [-0.37, 0.27] (227)	-0.30 [-0.64, 0.03] (208)	-0.66 [-0.96, -0.35] (156)
	swing $\geq 2\text{¢}$	+0.05 [-0.30, 0.47] (197)	-0.25 [-0.65, 0.23] (178)	-0.62 [-1.03, -0.18] (125)
	swing $\geq 4\text{¢}$	+0.55 [0.09, 1.14] (130)	+0.06 [-0.54, 0.79] (119)	-0.42 [-0.92, 0.39] (72)
	swing $\geq 6\text{¢}$	+0.32 [-0.46, 1.51] (69)	-0.21 [-1.31, 1.17] (56)	-0.45 [-0.94, 0.07] (29)

Notes: seed 0; stamp lag 2.0 s pre-registered, 3.0 s an optimistic sensitivity. Source: results/replay/selective/selective.json, research/replay/SELECTIVE.md.

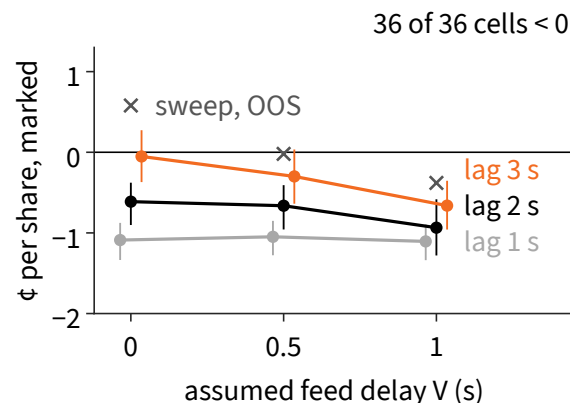


Figure A3. Replayed on 9 matches' real order books (recorded 2026-10-03), the 1 s trader loses at every stamp lag and feed delay. Every point called; net ¢ per share marked at +30 s, match-clustered 95% CI; x: the sweep's OOS curve. One day; assumed feed latency (licensed feed not purchased); paper only. Source: results/replay/replay.json.

Table A4. Factor regressions: v_2 's daily return has no meaningful equity-factor exposure. Fama–French three factors plus momentum, Newey–West 5 lags.

Specification	α %/day (t)	Mkt (t)	SMB (t)	HML (t)	Mom (t)	R^2	days
IS only (headline)	0.80 (8.5)	0.17 (1.4)	0.08 (0.5)	0.04 (0.4)	0.03 (0.4)	3.3%	138
IS, calendar days	0.68 (9.5)	–	–	–	–	2.3%	206
File (5 OOS days)	0.79 (8.5)	0.17 (1.3)	0.09 (0.5)	0.06 (0.6)	0.03 (0.5)	3.3%	142

Notes: the committed file includes five burned-OOS weekdays and is shown only for completeness. Source: results/alpha/alpha.json::C_factor_neutral; French (2026).

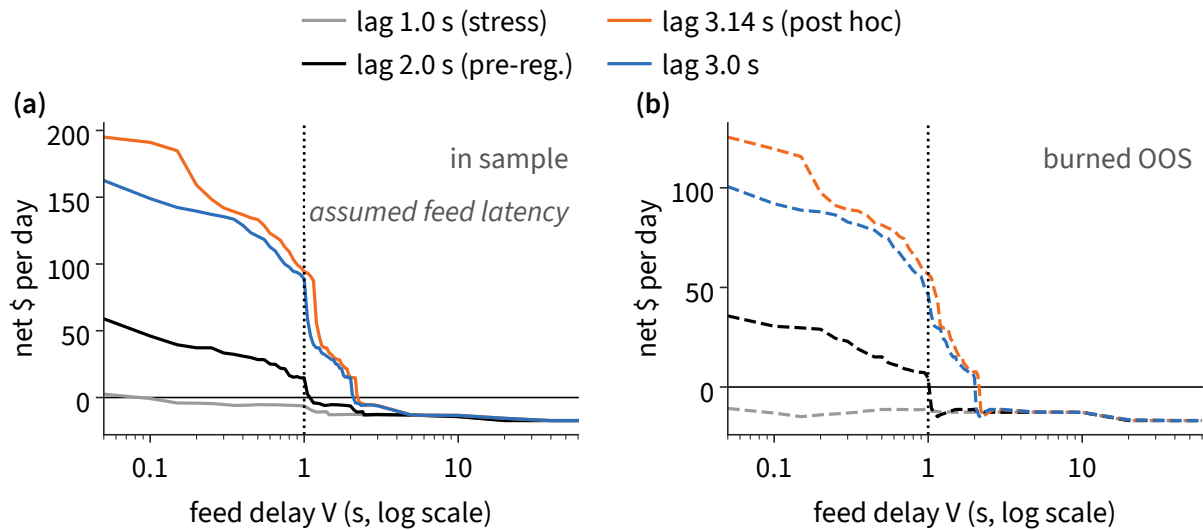


Figure A4. The break-even shifts one for one with the stamp lag. Net \$/day against feed delay for every stamp-lag reading, (a) IS and (b) burned OOS. Simulated: assumed feed latency (licensed feed not purchased); parameters measured. Source: results/tier0/latency_sweep.csv.

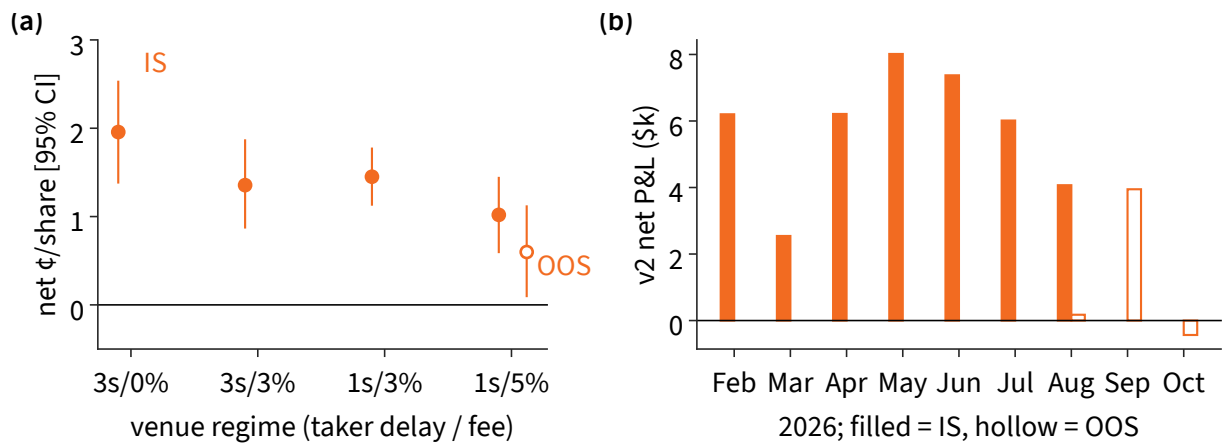


Figure A5. v2's edge falls as the venue's protection falls, and stays positive in every in-sample regime. (a) Net \$ per share by venue regime (taker delay/fee) with 95% CIs; (b) net P&L by month. Measured at the fast tier's own fills: the opportunity at their speed, not our execution. Source: results/risk/risk_stats.json.

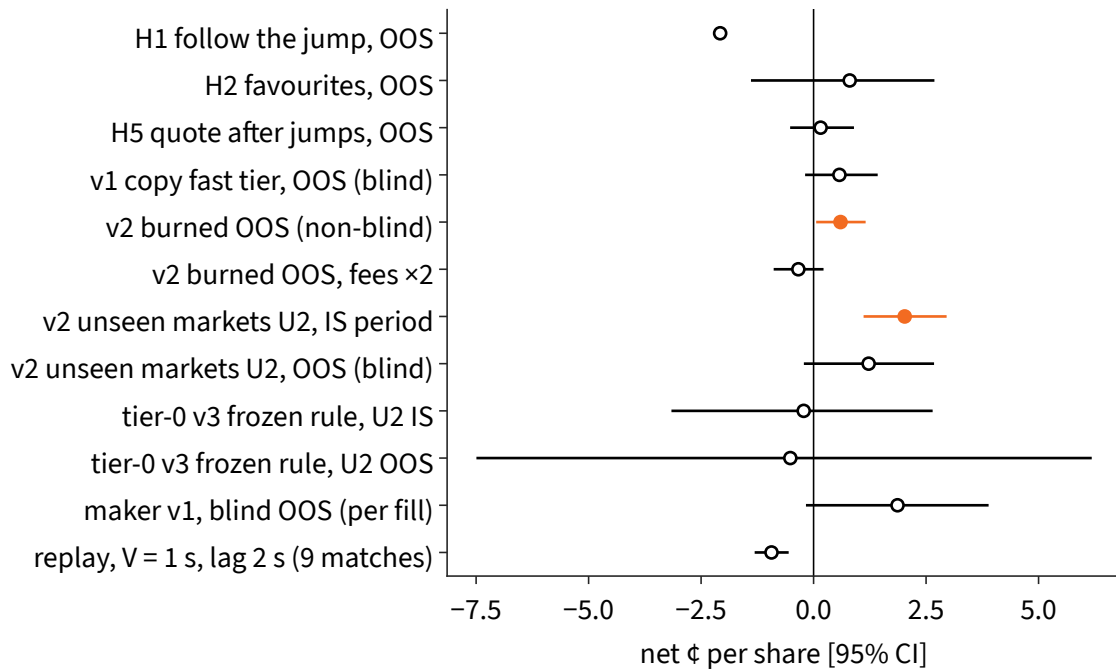


Figure A6. Every test on one axis. Point estimates and 95% CIs, net ¢ per share; orange = CI above zero, hollow = the test failed or has no edge. Source: as Table 3.

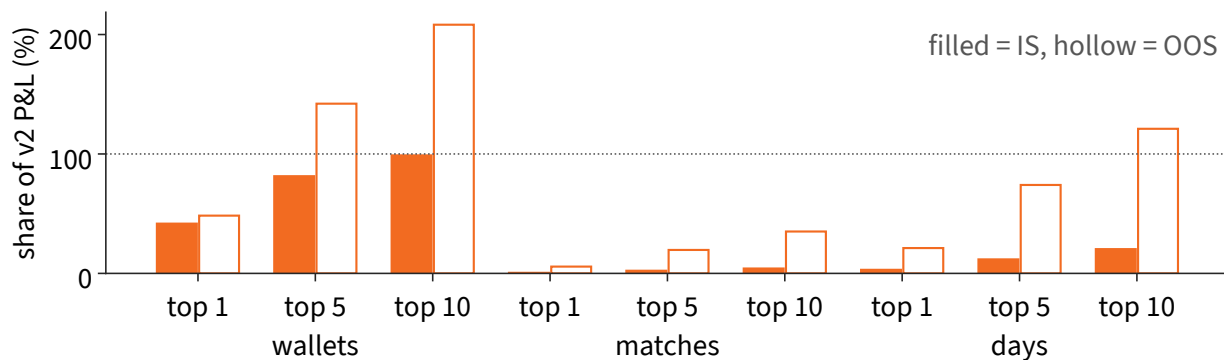


Figure A7. v2 P&L is concentrated in a few copied wallets. Share of v2 P&L carried by the top 1, 5 and 10 wallets, matches and days (above 100% means the rest lost money). Source: results/alpha/alpha.json.

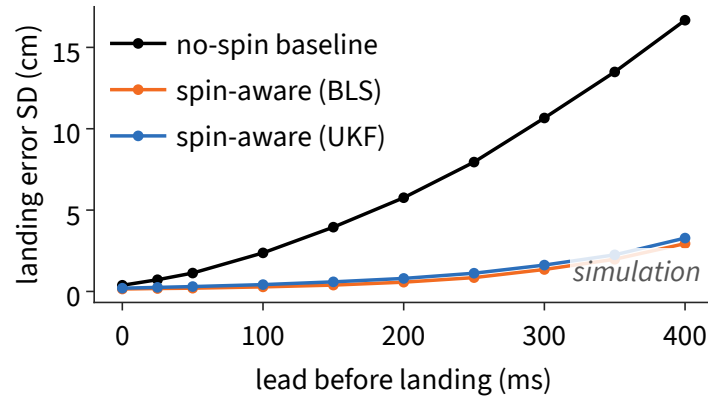


Figure A8. *Simulation: a spin-aware tennis filter cuts landing error.* Landing-point error against lead before the bounce for a 340 fps, 3.6 mm-noise ball tracker (simulation; no tennis video was used). At 200 ms: 5.76 cm without spin, 0.57 cm with it. Source: results/spin/tennis/key_numbers.json.

Table A5. *Variant tally: 4,219 configurations tried; the sensitivities chose nothing.*

Family	Count	What it chose
H1-H6 pre-registered and H5/H6 rules	44	frozen primaries (DEVIATIONS D6)
v2 lenses (exit, sizing, selection, cross-market, latency, Kalshi)	3,342	v2 (G_50pct_net100), IS only
v2-safe grid	24	v2-safe (secondary)
tier-0 pre-registered scenarios	432	none (sensitivity grid)
tier-0 v3 in-sample grid	360	frozen v3 rule (blind: FAIL)
maker v1 evaluated variants	8	maker v1 primary (blind: FAIL)
cost-stress cases	4	none (stress)
table-tennis tests TT1-TT5	5	none (untestable)
Total	4,219	
<i>Sensitivity, chose nothing: latency-sweep cells</i>	944	none
<i>Sensitivity, chose nothing: replay cells</i>	36	none

Source: results/paper/variants.json (each count's source file is listed there).

Table A6. Fixed-cost assumptions behind Fig. A2b (\$ per month unless stated).

Item	Low	Central	High	Label
Official live ATP/WTa point-data feed licence	\$1,250	\$5,000	\$10,000	ASSUMPTION
London gateway VPS (AWS eu-west-2, on-demand Linux)	\$34	\$77	\$98	ESTIMATE
Polymarket CLOB / data API	\$0	\$0	\$0	ESTIMATE
HiPerGator (training, backtests)	\$0	\$0	\$0	ESTIMATE
Courtside camera operator (tier-0 only)	\$36	\$36	\$36	ESTIMATE
Cloud GPU for CV inference, per covered event hour (tier-0 only)	\$1	\$1	\$1	ESTIMATE
Operator + GPU hours per covered match (tier-0 only)	\$3	\$4	\$5	ASSUMPTION
Camera position / venue access fee per event (tier-0 only)	\$0	\$250	\$500	ASSUMPTION
120 fps camera kit, amortised over 24 months (tier-0 only)	\$648	\$1,000	\$2,000	ESTIMATE

Notes: no public price exists for an official low-latency tennis feed; the range is anchored to reported figures (bases and URLs in the source file). Source: results/financials/financials.json::cost_assumptions.

Table A7. The latency ladder: method and source of every row of Fig. A1a.

Source	Delay	Method and citation
Our CV engine, call-ready (p50–p99)	4.6–12.2 ms	measured, one L4 GPU, 120 fps held-out video
Our WebRTC → CV pipeline	46 ms p50, 67 ms p99	measured: our own clip over loopback, frame send to call (55 calls; engine fed below real time so it keeps up); not a match feed
Network, Florida → venue gateway	67 ms	measured (live feed p50 65 ms)
In-venue camera	≤0.15 s	physical bound; not available to us
Venue taker order delay	1 s	venue rule (Polymarket, 2026a)
Official umpire stamp after the bounce	1–3 s	unmeasured model grid (Regen Sports, 2026; Stats Perform, nd)
Book reprice vs official stamp	–1.2 s	measured median, $n = 482$
Licensed betting video	0.5–8 s	vendor claims, unverified, none for tennis (Stats Perform, nd; NEXT.io, nd; Sportradar, nd; Ably, nd)
Fastest public stream of a listed match	12.2 s	measured, preliminary (no YouTube-derived figure used)
ESPN / PM sports / WTA vs book	28.2 / 30.0 / 44.1 s	measured medians, 2026-10-03 (ESPN, 2026; WTA, 2026)
TV and public streams, general	0.9–60 s	published comparisons (ISPreview, 2021; GL Systemhaus, 2018; The Desk, 2026)

Source: research/v2/latency/results.json, results/tier0/latency_sweep.json::sources, results/home_stream/sub_second_routes.json, results/engine/online_vs_offline.json.

Table A8. Every logged read of held-out data (76 lines of results/oos_peek.log at build time), with its class.

UTC	Class	What was read (truncated)
10-03 10:42	other	OOS prints built
10-03 13:24	non-blind	v2 evaluated on burned OOS (non-blind, labelled)
10-03 13:53	non-blind	v2-causal (D9 fix) evaluated on burned OOS (non-blind)
10-03 17:23	non-blind	tier0 counterfactual: OOS jump table built (burned OOS, non-blind)
10-03 17:29	non-blind	tier0 counterfactual grid evaluated on burned OOS (non-blind, labelled)
10-03 17:31	blind first run	U2 out-of-universe test evaluated (PREREG research/v2/expand, first run)
10-03 17:37	non-blind	v2-safe (lowloss PREREG) evaluated on burned OOS (non-blind, labelled)
10-03 17:37	blind first run	v2-safe U2 out-of-universe test evaluated (PREREG research/v2/lowloss, first run)
10-03 17:43	audit	tier0 independent rebuild: OOS prints read for jump onsets + stale price (burned OOS, non-blind, labelled)
10-03 17:45	audit	tier0 independent rebuild: OOS prints read for jump onsets + stale price (burned OOS, non-blind, labelled)
10-03 17:48	audit	tier0 independent rebuild: OOS prints read for jump onsets + stale price (burned OOS, non-blind, labelled)
10-03 18:00	non-blind	v2 cost stress (fees/costs doubled, book fixed) on burned OOS (non-blind, no parameter change)
10-03 18:06	non-blind	tier0 corrections: OOS post-jump prices built (burned OOS, non-blind, labelled)
10-03 18:13	non-blind	tier0 counterfactual (verifier corrections) evaluated on burned OOS (non-blind, labelled)
10-03 18:27	descriptive	maker venue rules: descriptive in-play book spread/depth + tick changes read from data/live_v2 forward recordings (no strategy, no P&L, no parameter c...
10-03 18:42	non-blind	financials: v2 / v2-safe size-scaled variants (0.5x, 2x, 5x, all prints) evaluated on burned OOS (non-blind, capacity curve only)
10-03 18:47	descriptive	maker live engine plumbing: data/live_v2 clob_17/clob_13 read for message ordering only (price_change vs last_trade_price timestamps, trade dedupe); n...
10-03 18:49	blind first run	maker v1 blind OOS evaluated (first run) (PREREG research/v2/maker/PREREG.md (A); first use of OOS side-market tapes; burned OOS moneyline tapes used ...
10-03 18:49	audit	maker v1 OOS reproduction (recompute committed results.json; no parameter change)
10-03 19:00	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_11.jsonl.gz,data/live_v2/clob_20261003_1123_2026100...
10-03 19:13	live	maker v1 live paper TEST run (not the pre-registered session) START: run=20261003T190317Z commit=56a9c8f77b1aaf7b1306c2f1b06d6a2e05c798a8+dirty until=...

UTC	Class	What was read (truncated)
10-03 19:15	audit	P17 clean-clone reproduce.sh (HEAD 56a9c8f, scratch clone, data/linked, backtest workers=1): re-runs the committed frozen scripts incl. run_all.py --...
10-03 19:19	live	maker v1 live paper engine REPLAY (B2: own raw log) on data/live_maker/raw_20261003T190317Z.jsonl.gz window=-+Nonemin clock=recv: run=20261003T191913Z...
10-03 19:20	non-blind	PM review P07: wallet-clustered bootstrap CI of the frozen v2 causal book on burned OOS (non-blind, labelled; the CI the note already quotes, no rule ...
10-03 19:23	blind first run	table tennis TT1-TT4 evaluated (first run)
10-03 19:25	descriptive	TT5 signal decay (scripts/signal_decay.py, commit a5769c786f935dcc05ff2ab90a1947ad0cde4cb5+dirty(signal_decay.py)): reads table tennis UTT OOS prints ...
10-03 19:28	other	table tennis post-hoc diagnostics after TT1-TT4 (closing-print size, fast-tier distance; not pre-registered, no verdict change)
10-03 19:28	live	maker v1 live paper TEST run (not the pre-registered session) START: run=20261003T191915Z commit=ad164da6da24325afcfdbab8fe24dcf5f0516fed+dirty until=...
10-03 19:58	live	maker v1 live paper session (B) START: run=20261003T193534Z commit=114c722e6b203421d8b4f55a5da2b54a1658b658 until=2026-10-04T11:30:00.000Z (PREREG res...
10-03 20:02	audit	audit: independent TT3 reproduction (research/tt/audit_tt3.py; reads UTT IS+OOS tapes/prints, already evaluated in 2195e86) plus labelled post-hoc dia...
10-03 20:03	audit	audit: table tennis checks (research/tt/audit_checks.py): provenance, fees/delays, voids, counts, liquidity; reproduces TT1 closing bins (IS+OOS, alre...
10-03 20:04	audit	audit: financials/risk audit (scripts/financials_audit.py) re-reads the frozen v2 burned-OOS trades and maker blind-OOS fills to reproduce numbers alr...
10-03 20:07	audit	audit: RE-RUN of research/tt/audit_tt3.py with three added post-hoc counts (closest wallet's share of TT2 others' prints, 0-3 s prints at fav >= 0.95 ...
10-03 20:07	audit	audit: RE-RUN of research/tt/audit_tt3.py (traceback location recorded as repo-relative src/v2.py frames only; same reads and numbers, no rule or verd...
10-03 20:09	descriptive	alpha breakdown: descriptive splits of existing v2 trades incl. burned OOS (no parameter change)
10-03 20:10	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[12].jsonl.gz window=2026-10-03T12:00:00.000Z+60.0...
10-03 20:10	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[12].jsonl.gz window=2026-10-03T12:00:00.000Z+60.0...
10-03 20:11	audit	audit: maker v1 blind OOS (A) independent recompute (own code in scratchpad, no import of scripts/maker_oos.py or research/v2/maker/oos_*.py); reads O...

UTC	Class	What was read (truncated)
10-03 20:13	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[123].jsonl.gz window=2026-10-03T13:00:00.000Z+60....
10-03 20:13	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[1234].jsonl.gz window=2026-10-03T14:00:00.000Z+12...
10-03 20:14	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_11.jsonl.gz window=2026-10-03T11:35:00.000Z+25.0min...
10-03 20:14	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[56789].jsonl.gz window=2026-10-03T16:00:00.000Z+2...
10-03 20:16	non-blind	tier0 latency sweep on burned OOS (non-blind, labelled; no parameter chosen)
10-03 20:16	audit	audit: TT5 same-block split and v2 trade join (research/decay/audit_sameblock.py): re-reads tennis U1 IS + burned OOS prints (non-blind, already evalu...
10-03 20:18	audit	audit: selection check for maker v1 OOS (A): data-api /trades counts and notional (no prices used, no signal, no P&L) for a random sample of 60 OOS si...
10-03 20:21	audit	audit: maker v1 OOS (A) stress (viii) alternative fee reading (makerBaseFee 1000 bps x min(px,1-px), CTF-exchange form) on the already-evaluated book...
10-03 20:22	audit	audit: provenance spot-check of maker v1 OOS cache: re-download 8 cached OOS side tapes (data-api) and 1 Gamma events batch and compare with the cache...
10-03 20:22	live	maker v1 live paper engine REPLAY (B2: own raw log) on data/live_maker/raw_20261003T191915Z.jsonl.gz window=-+Nonemin clock=recv: run=20261003T202228Z...
10-03 20:22	live	maker v1 live paper engine REPLAY (B2: own raw log) on /private/tmp/claude-501/-Users-ojasvamishra32-Library-Application-Support-Claude-scratch-worksp...
10-03 20:24	non-blind	tier0 latency sweep on burned OOS (non-blind, labelled; no parameter chosen)
10-03 20:25	audit	tier0 v3 blind test (a) burned OOS START (scripts/tier0_v3_blind.py; PREREG research/v2/tier0_v3/PREREG.md section 5; non-blind window seen by v1/v2/t...
10-03 20:26	blind first run	tier0 v3 blind test (b) U2 START (scripts/tier0_v3_blind.py; PREREG research/v2/tier0_v3/PREREG.md section 4; first tier-0 table on U2: jump tables + ...
10-03 20:28	blind first run	tier0 v3 blind tests (a)+(b) DONE: burned OOS FAIL, U2-IS FAIL, U2-OOS FAIL -> results/tier0_v3/blind.json
10-03 20:29	blind first run	tier0 v3 blind tests (a)+(b) DONE: burned OOS FAIL, U2-IS FAIL, U2-OOS FAIL -> results/tier0_v3/blind.json
10-03 20:37	audit	audit: tier-0 v3 blind tests adversarial audit (independent code in scratchpad; COUNTERFACTUAL): re-reads saved v3 burned-OOS/U2 outputs (results/tier...

UTC	Class	What was read (truncated)
10-03 20:40	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[123].jsonl.gz window=2026-10-03T13:00:00.000Z+60....
10-03 20:45	audit	independent rebuild: tier-0 frozen v3 re-implemented in scratchpad (no scripts/tier0_v3_* import; COUNTERFACTUAL) reads U2 prints/universe/raw tapes, ...
10-03 20:46	live	maker v1 live paper engine REPLAY (recorded live data) on data/live_v2/clob_20261003_1123_20261003_1[123].jsonl.gz window=2026-10-03T13:00:00.000Z+60....
10-03 20:52	live	maker v1 live paper session (B) DEVIATION L11-L14 (research/v2/maker/DEVIATIONS_LIVE.md): session run=20261003T193534Z PID 65274 stopped by SIGTERM at...
10-03 21:05	descriptive	alpha breakdown: descriptive splits of existing v2 trades incl. burned OOS (no parameter change)
10-03 20:17	audit	tier0 latency sweep on burned OOS (non-blind, labelled; no parameter chosen) [HiPerGator job 44606812; line written to the cluster copy of this log, s...
10-03 20:36	audit	tier0 latency sweep on burned OOS (non-blind, labelled; no parameter chosen) [HiPerGator job 44608194, the run behind commit f5ff9ff; line written to ...
10-03 20:53	audit	tier0 latency sweep on burned OOS (non-blind, labelled; no parameter chosen) [audit: independent recompute of sweep cells incl. burned OOS video V = 1...
10-03 20:59	audit	tier0 latency sweep on burned OOS (non-blind, labelled; no parameter chosen) [HiPerGator job 44610737, audit re-run adding stamp-lag cells; line writt...
10-03 21:14	live	match replay: forward live recordings of 2026-10-03 used for a labelled 1 s-feed counterfactual replay (no parameter chosen) [logged before the replay...
10-03 21:56	live	match replay: forward live recordings of 2026-10-03 used for a labelled 1 s-feed counterfactual replay (no parameter chosen) [visuals: scripts/match_r...
10-03 22:18	non-blind	capacity study: burned OOS size/cap/coverage grid (non-blind, no parameter chosen) [scripts/capacity_study.py: local v2 1x check, then the HiPerGator ...
10-03 22:24	live	match replay: forward live recordings of 2026-10-03 used for a labelled 1 s-feed counterfactual replay (no parameter chosen) [audit: scripts/match_rep...
10-03 23:01	live	match replay selective variant: exploratory, post hoc (forward live recordings of 2026-10-03) [scripts/match_replay_selective.py: the replay restricte...
10-03 23:17	audit	capacity study: burned OOS size/cap/coverage grid (non-blind, no parameter chosen) [audit of commit f9d8f94: independent recompute of 5 grid cells inc...
10-03 23:18	non-blind	redteam causal-CV cell (INTEGRATION_TODO C5; scripts/redteam_causal_cv.py): own120_engine = live causal engine recall/precision at V 0-3 s under tourn...

UTC	Class	What was read (truncated)
10-03 23:23	non-blind	backtest example match (video asset): latency-sweep rerun at V=1 s, calibrated reading, read burned OOS prints for per-match P&L (non-blind, illustrat...
10-03 23:24	live	match replay selective variant: exploratory, post hoc (forward live recordings of 2026-10-03) [audit of commit b20a4e4: adversarial check of scripts/m...
10-04 00:28	non-blind	capacity study: burned OOS size/cap/coverage grid (non-blind, no parameter chosen) [fix pass after the capacity and e2e verifiers: pre-flight test of ...
10-04 00:28	audit	capacity study: burned OOS size/cap/coverage grid (non-blind, no parameter chosen) [scripts/capacity_study.py --grid on Ojasvas-MacBook-Air-3.local, r...
10-04 00:36	audit	capacity study: burned OOS size/cap/coverage grid (non-blind, no parameter chosen) [scripts/capacity_study.py --grid --resume on Ojasvas-MacBook-Air-3...

Table A9. *Pre-registration and deviation trail (git history of the hypothesis, protocol and deviation files).*

Commit	UTC	Subject
7232986	2026-10-03 09:50	Pre-register hypotheses before any results
0f1cc89	2026-10-03 10:15	Data pipeline, Markov model, physics sim, strategies, tier and fast-tier analysis
9d61d1b	2026-10-03 10:32	Freeze all strategy settings before out-of-sample evaluation
69a675f	2026-10-03 10:48	Out-of-sample evaluation, diagnostics, figure fixes
be64ea4	2026-10-03 12:00	Ball tracking results (table tennis H3, tennis broadcast), interactive visual, live-book paper engine
1f229bc	2026-10-03 13:22	Pre-register v2 rule and forward test; add v2 research lenses and verifications
845666f	2026-10-03 13:55	Fix onset hindsight: causal v2; amend forward-test pre-registration before any forward data
e7f34d8	2026-10-03 13:56	Forward window starts 14:00 UTC, after the amended pre-registration
8ede061	2026-10-03 15:00	Pitch deck (12 slides, reviewed); note framing fixes; D10 corrections
337bf7c	2026-10-03 17:12	Pre-register out-of-universe test of frozen v2 (U2)
342d515	2026-10-03 17:30	Pre-register v2-safe (loss-averse, chosen on IS only)
c29734b	2026-10-03 18:26	Pre-register maker v1 blind OOS test and live paper session
0f01362	2026-10-03 18:31	Pre-register table tennis study (TT1-TT5) before fetching tapes
a5769c7	2026-10-03 19:24	Tier-0 CV-edge counterfactual (assumed licensed feed + courtside camera; measured parameters): pre-registration, original run, two verifier reports, revision with live-book fill pricing; original kept as record
ba0c3d4	2026-10-03 20:16	Pre-register frozen tier-0 v3 and its blind tests
0e083ed	2026-10-03 21:14	Match replay protocol: selection rule and per-point procedure, written before any P&L